

The Switchback Law

The prime-angle sign sequence is scale-invariant — the same seesaw in the hundreds and the millions

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What this is. An empirical regularity with a clean lens and one open thread. The Prime-Triangle angle wobbles under its 45° ceiling; strip away the *size* of the wobble and keep only the **sign** of each change (up or down), and that sign-sequence — how often it flips, how long same-direction runs last — is **the same at every scale**, even as the wobble itself shrinks by orders of magnitude. It also yields a usable **betting rule** for the next sign, and its fine structure is the wheel. Tags: **[emp]** measured · **[null]** shuffle-controlled · **[interp]** placement · **[open]** the mod6 thread. Companion: *The Prime-Triangle Angle* (the wobble and its ceiling); *Prime-Gap Memory and the Differencing Trap* (the two-null discipline used here).

*Placement up front (no inflation): this is a fresh, visual face of something number theory already expects — the **scale-invariance of the normalized local statistics of primes** (the Cramér / Hardy–Littlewood picture). It is not a new theorem. Its value is the lens (it makes that invariance graspable and turns it into a rule you can use) and the mod6 fine structure (§5). It sits alongside the other shadows in this collection, not above them.*

1. The object: switchbacks

For consecutive primes, the Prime-Triangle angle is $\alpha_n = \arctan(p_n/p_{n+1}) \approx 45^\circ - \frac{90}{\pi} g_n/p_{n+1}$ — a rescaled gap, pinned just under 45° . Its **change** $\Delta\alpha_n = \alpha_n - \alpha_{n-1}$ carries a sign: **up** (the gap shrank, the angle rose toward the ceiling) or **down** (the gap grew). Reading only that sign, the sequence is a string of switchbacks: $- - + - + + - \dots$

On the ordinary number line this sign-structure is invisible — “this gap, then subtract that gap, then the next” has no picture. The triangle is what renders it; that is why the pattern is *seeable* at all.

2. The run-length law

Same-direction runs are short, and they die **faster than a coin**. Over the primes in $[10^6, 3 \times 10^6]$:

run of length	share of runs	a fair coin would give
1	63.1%	50%
2	27.3%	25%
3	7.7%	12.5%
4	1.6%	6.25%
5	0.3%	3.1%
6–7	0.07%	1.6%

[emp] Each extra same-sign step is $\sim 4\text{--}5\times$ less likely than the last — the sequence *over-alternates* relative to random. Equivalently, the **reversal probability climbs with run length**:

after a run of...	the next change flips
1	63%
2	74%
3	80%
4	82%

So the sign is genuinely **predictable**: once you have seen two or three moves the same way, bet on a reversal and you are right three or four times out of five. (This is the discipline the wobble note’s “runs $\approx 1/3$ ” constant pointed at, now read as a usable table.)

3. The law is scale-invariant — the headline

The *amplitude* of the wobble collapses with scale — $\text{RMS}(\Delta\alpha) \approx 33 \ln p/p$, four orders of magnitude smaller near 10^6 than near 10^2 . The **sign-structure does not move at all**:

run length	1	2	3	4	5	6
hundreds (first 1000 primes)	62.2%	29.5%	6.9%	1.3%	—	—
millions ($[10^6, 3 \times 10^6]$)	63.1%	27.3%	7.7%	1.6%	0.3%	0.06%

The two distributions are the same. In the first thousand primes there are **exactly nine** runs of length four and none longer; in the millions the identical *rate* produces occasional 5s, 6s, a rare 7 — so “four is the maximum” is a sample-size floor, not a hard ceiling, but the **law generating it is fixed across scale**. Zoom into the millions and the switchbacks look exactly like the hundreds. *This is the point of the note.*

4. Why the runs die fast: the ceiling, felt from the changes

The over-alternation is the boundedness of the angle seen from the change side. Because primes increase with bounded gaps, α is trapped in a shrinking band below 45° ; it cannot drift, so its changes cannot run the same way for long without being pulled back — an **overcorrection** that always returns the angle to the band (and, being a return, tends to overshoot, seeding the next reversal). That this is structural and not generic noise is shown in the companion note by the two-null test: shuffle the *gaps* and the band-and-cancellation survive; shuffle the *changes* and let them free-float, and the reconstructed angle wanders straight through 45° . The switchback tightness is held in place by the primes being a real monotone, bounded-gap sequence.

5. The fine structure is the wheel: a mod6 suppression

Within the sign-law lives an arithmetic fingerprint. Look at the size of the gap-change, $|\Delta g| = |g_{n+1} - g_n|$, over $[10^6, 3 \times 10^6]$:

$ \Delta g $	2	4	6	8	10	12	14
frequency	16.8%	13.3%	5.1%	11.4%	8.8%	3.0%	6.6%

[**emp**] The multiples of 6 are **suppressed** — sharp local minima at 6 and 12. Since gaps are even, $|\Delta g| \equiv 0 \pmod{6}$ means two consecutive gaps are *congruent mod6*; the dip says **consecutive gaps avoid repeating their residue mod6**. That is a Lemke Oliver–Soundararajan-type bias (the “unexpected biases in consecutive primes,” 2016) — the same primordial wheel that casts every other shadow in this collection, here reading off the *changes* between gaps. [**open**] Whether the exact depth of the 6- and 12-dips matches the singular-series prediction is the one genuinely open thread here, and the natural next computation.

6. Where it lands, and what it’s good for

Prediction. The betting table of §2 is scale-invariant (§3), so it *transfers*: hunting the sign-rhythm around a record prime, the wobble sizes up there are meaningless, but the run-length odds you learned in the hundreds are exactly the odds in the billions. Combined with the Seven-Sisters generator ($2p + k$ for the wheel-strong

offsets), this is the honest content of “how to guess where the next prime leans” — not a formula, a *rhythm with known odds*.

Placement, plainly. The switchback law is a real, clean, scale-invariant regularity — but it is a *consequence* of the conjectured scale-invariance of normalized prime local statistics, not a new fact about the deep structure of primes. Its contribution is the **lens** (it makes that invariance visible and usable, through the angle sign-sequence that the number line hides) and the mod6 **fine structure** (§5), which ties it back to the wheel. It is one more shadow — the seesaw one — not a summit.

Appendix. Reproduction

```
import math, bisect
from collections import Counter, defaultdict
def sieve(n):
    b=bytearray([1]*(n+1)); b[0]=b[1]=0; i=2
    while i*i<=n:
        if b[i]: b[i*i::i]=bytearray(len(b[i*i::i]))
        i+=1
    return b
B=sieve(3_000_000); allp=[i for i in range(2,3_000_000) if B[i]]
def runs_and_reversal(P):
    a=[math.degrees(math.atan(P[i]/P[i+1])) for i in range(len(P)-1)]
    da=[a[i+1]-a[i] for i in range(len(a)-1)]
    s=[1 if d>0 else -1 if d<0 else 0 for d in da]
    runlen=Counter(); i=0
    while i<len(s):
        j=i
        while j<len(s) and s[j]==s[i] and s[i]!=0: j+=1
        if s[i]!=0: runlen[j-i]+=1
        i=max(j,i+1)
    return runlen
i1=bisect.bisect_left(allp,1_000_000); i2=bisect.bisect_left(allp,3_000_000)
print('hundreds:', runs_and_reversal(allp[:1000]))
print('millions:', runs_and_reversal(allp[i1:i2]))
# mod-6 suppression:
g=[allp[i+1]-allp[i] for i in range(i1,i2)]
print(Counter(abs(g[i+1]-g[i]) for i in range(len(g)-1)).most_common(8))
```

References

- A. Proxmire, *The Prime-Triangle Angle* (this collection) — the wobble, the 45° ceiling, the $33 \ln p/p$ envelope.
- A. Proxmire, *Prime-Gap Memory and the Differencing Trap* (this collection) — the two-null discipline (shuffle gaps vs shuffle changes) used in §4.
- R. J. Lemke Oliver, K. Soundararajan, *Unexpected biases in the distribution of consecutive primes* (PNAS, 2016) — the residue-correlation mechanism behind the mod6 suppression of §5.
- Empirical basis: *Prime Trigonometry Data* (author’s spreadsheet, first 1000 pairs) cross-checked against a sieve to 3×10^6 .